

rray Logical Pogram contd...

```
package com.ravi.one_dimensional_basic;

import java.util.Arrays;

/**Find the Second largest element without duplicate

public class ArrayDemo17 {

    public static void main(String[] args)
    {
        int []arr = {12,90,56,34,77};
        IO.println("Original Array :"+Arrays.toString(arr));

        //Descending sorting logic
        int temp = 0;

        for(int i=0; i<arr.length; i++)
        {
            for(int j=i+1; j<arr.length; j++)
            {
                if(arr[i] < arr[j])
                {
                    temp = arr[i];
                    arr[i] = arr[j];
                    arr[j] = temp;
                }
            }
        }

        IO.println("Array elements in descending order :"+Arrays.toString(arr));
        IO.println("Second largest eleemnt is :"+arr[1]);
    }
}
```

```
package com.ravi.one_dimensional_basic;

import java.util.Arrays;

/**Find Second largest element with duplicate

public class ArrayDemo18 {

    public static void main(String[] args)
    {
        int []arr = {90, 12,90, 46, 56,90,77, 77}; //12 46 56 77 77 90 90 90
        IO.println("Original Array :"+Arrays.toString(arr));

        //Sort the Array in ascending order
        int temp = 0;
        for(int i=0; i<arr.length; i++)
        {
            for(int j=i+1; j<arr.length; j++)
            {
                if(arr[i] > arr[j])
                {
                    temp = arr[i];
                    arr[i] = arr[j];
                    arr[j] = temp;
                }
            }
        }

        //12 46 56 77 77 90 90 90

        int largest = arr[arr.length-1];
        int secondLargest = Integer.MIN_VALUE; //[Integer.MIN_VALUE OR -1]

        //for loop to find the 2nd largest element
        for(int i=arr.length-2; i>=0; i--)
        {
            if(arr[i] !=largest) //Will by pass duplicate
            {
                secondLargest = arr[i];
                break; //break is compulsory
            }
        }

        IO.println("second largest element is :"+secondLargest);
    }
}
```

```
package com.ravi.one_dimensional_basic;

/*
 * Print the index position of two array elements whose sum is equal to 9
 * Two sum problem
 * int[] arr = {2, 7, 6, 11, 15, 3};
 * int target = 9;
 */

public class ArrayDemo19
{
    void main()
    {
        int[] arr = {2, 7, 6, 11, 15, 3};
        int target = 9;

        for(int i=0; i<arr.length; i++)
        {
            for(int j=i+1; j<arr.length; j++)
            {
                if(arr[i]+arr[j] == target)
                {
                    IO.println(target+" found at "+i+", "+j+" index position");
                }
            }
        }
    }
}
```

```
package com.ravi.one_dimensional_basic;

/*Given an array of n integers, return true if the array is sorted in
non-decreasing order or not */

public class ArrayDemo20
{
    public static void main(String[] args)
    {
        int[] values = {1, 3, 5, 5, 8, 9, 10, 5};
        IO.println("Is array elemnets are sorted in ascending order :");
        IO.println(isSortedInAscending(values));
    }

    public static boolean isSortedInAscending(int []arr)
    {
        for(int i=0; i<arr.length-1; i++) //length = 7
        {
            if(arr[i] > arr[i+1]) //10 > 9
            {
                return false;
            }
        }

        return true;
    }
}
```

```
package com.ravi.one_dimensional_basic;

import java.util.Arrays;

/*
 * Given an array of N integers, left rotate the array by one place.?
 * Input : 1,2,3,4,5
 * Output :2 3 4 5 1
 */

public class ArrayDemo21 {

    public static void main(String[] args)
    {
        int []arr = {1,2,3,4,5};
        IO.println("Original Array :"+Arrays.toString(arr));

        int temp = arr[0]; //temp = 1

        for(int i=0; i<arr.length-1; i++)
        {
            arr[i] = arr[i+1];
        }

        arr[arr.length-1] = temp;

        IO.println("Array element after rotating 1 place :"+ Arrays.toString(arr));
    }
}
```

Assigenmnt : Rotate the array for n number of position using method

```
package com.ravi.one_dimensional_basic;

import java.util.Arrays;

/*
 *** You are given an array of integers, your task is to move all the zeros in
the array to the end.
Input : [0, 1, 3, 4, 0, 9, 5]
Output : [1, 3, 4, 9, 5, 0, 0]
 */

public class ArrayDemo22 {

    public static void main(String[] args)
    {
        int []arr = {0, 1, 3, 4, 0, 9, 5};
        IO.println("Original Array "+Arrays.toString(arr));
        int temp = 0;
        for(int i=0; i<arr.length; i++)
        {
            if(arr[i]==0)
            {
                for(int j=i+1; j<arr.length; j++)
                {
                    if(arr[j]!=0)
                    {
                        //Swapping
                        temp = arr[i];
                        arr[i] = arr[j];
                        arr[j] = temp;
                        break;
                    }
                }
            }
        }

        IO.println("Array elements after moving all the zeros :"+Arrays.toString(arr));
    }
}
```

```
package com.ravi.one_dimensional_basic;

import java.util.Arrays;

/*
 * Q23) Given an array that contains only 1 and 0 return the count of maximum
 * consecutive ones in the array?
 * Input : 1,1,0,1,1,1,0,1,1,1,1,1
 * Output : Maximum Consecutive Ones: 5
 */

public class ArrayDemo23 {

    public static void main(String[] args)
    {
        int []arr = {1,1,0,1,1,1,1,1,1,1,0,1,1,1,1,1};
        IO.println("Original Array :"+Arrays.toString(arr));

        int count = 0; //To count the number of Ones
        int maxCount = 0; //To Store the Consecutive Ones

        for(int i=0; i<arr.length; i++)
        {
            if(arr[i]==1)
            {
                count++; //2

                if(count > maxCount)
                {
                    maxCount = count;
                }
            }
            else
            {
                count = 0; //reset the value
            }
        }

        IO.println("Maximum Consecutive Ones :"+maxCount);
    }
}
```